

Basic Classification Model

A simple supervised learning project that trains a K-Nearest Neighbors classifier on the Iris flower dataset to predict species from flower measurements.

Requirements

- Python 3.8+
- `scikit-learn`
- `pandas`

Install dependencies:

```
pip install scikit-learn pandas
```

Concepts Demonstrated

Data handling, supervised learning basics, model training

What It Does

1. **Load and understand the data** — loads the built-in Iris dataset (150 samples, 4 features: sepal length/width, petal length/width, across 3 species) and previews it
2. **Split the data** — divides it into 80% training and 20% testing sets
3. **Train a classifier** — fits a K-Nearest Neighbors model ($k=3$) on the training data
4. **Evaluate** — predicts on the test set and prints accuracy plus a full classification report (precision, recall, f1-score per class)
5. **Demo prediction** — runs one sample through the trained model and compares the prediction to the actual label

How to Run

```
python classification_model.py
```

Example Output

Model accuracy: 100.00%

Classification report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11

Sample input: [6.1 2.8 4.7 1.2]

Predicted species: versicolor

Actual species: versicolor

Why KNN?

K-Nearest Neighbors classifies a new data point by looking at the “k” closest points in the training data and taking a majority vote of their labels. It’s one of the simplest classification algorithms to understand, making it a good starting point for learning supervised learning concepts.